

# Session 01 - What is TypeScript, TS vs JS, Setup, Compilation

Naveen AutomationLabs | 2026 Playwright Batch G1

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## 1. What is TypeScript?

TypeScript is a strongly typed programming language built on top of JavaScript. It was developed by Microsoft. TypeScript adds static typing, interfaces, enums, and other features to JavaScript.

- TypeScript is a superset of JavaScript -- every valid JS code is valid TS code.
- TypeScript code cannot run directly in browsers or Node.js. It must be compiled (transpiled) to JavaScript first.
- File extension: .ts (instead of .js)

## 2. TypeScript vs JavaScript

- JS is dynamically typed. TS is statically typed -- you declare types for variables and function parameters.
- JS catches type errors at runtime. TS catches them at compile time, before you even run the code.
- TS has interfaces, enums, generics, access modifiers -- JS does not have these natively.
- TS compiles down to plain JS. The browser or Node.js only runs the compiled JS output.

## 3. Installing TypeScript

Make sure Node.js is already installed on your machine.

### **Step 1: Install TypeScript globally**

```
npm install -g typescript
```

### **Step 2: Verify installation**

```
tsc --version
```

### **Step 3: Initialize a TypeScript project**

```
tsc --init
```

This creates a tsconfig.json file in your project folder. This file holds all TypeScript compiler settings.

## 4. Compilation - How TypeScript Runs

TypeScript does not run directly. You must compile .ts files into .js files first, then run the .js file.

### **Manual compilation:**

```
tsc filename.ts // compiles to filename.js  
node filename.js // run the compiled JS file
```

### ***Watch mode (auto-recompile on save):***

```
tsc --watch
```

In watch mode, every time you save a .ts file, it automatically recompiles to .js.

## **5. tsconfig.json - Key Settings**

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The tsconfig.json file controls how TypeScript compiles your code. Here are the most important settings:

- "target": "ES2016" -- which JS version to compile to.
- "module": "commonjs" -- module system to use.
- "outDir": "./dist" -- output folder for compiled JS files.
- "rootDir": "./src" -- folder where your .ts source files are.
- "strict": true -- enables all strict type-checking options.

```
{
  "compilerOptions": {
    "target": "ES2016",
    "module": "commonjs",
    "outDir": "./dist",
    "rootDir": "./src",
    "strict": true
  }
}
```

## **6. Why TypeScript for Playwright?**

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- Playwright officially recommends TypeScript for writing tests.
- You get auto-complete and IntelliSense in VS Code because of type information.
- Errors are caught early at compile time, not when the test is running.
- Interfaces and types make page objects and test data cleaner and more structured.
- All Playwright APIs have TypeScript type definitions built in.